MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	6	S					INDEX NUMBER		

H2 BIOLOGY 9744/03

Paper 3 Long Structured and Free-response Questions

18 September 2017
2 hours

Candidates answer on the Question Paper.
No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **ONE** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 24
2	/ 18
3	/ 8
Section B	
4 or 5	/ 25
Total	/ 75

This paper consists of **18** printed pages.

[Turn over]

Section A

Answer **all** the questions in this section.

QUESTION 1

Some types of snake kill their prey and defend themselves by means of a poisonous bite. Fangs (hollow teeth) inject venom from specialized glands into the victim. The venom contains a protein, which is a toxin.

Different species of snake have toxins that act in different ways. Hemolytic toxins are enzymes that hydrolyze phospholipids. They damage tissues, including heart muscle, which lead to cardiac arrest. Neurotoxins, such as the one produced by green mamba snakes, bind to receptor proteins on the surface membranes of nerve cells or muscle fibers. This interferes with the transmission of nerve impulse, leading to muscle paralysis and heart failures.

Fig. 1.1 shows the molecular structure of fasciculin-2, a neurotoxin produced by the green mamba snake.

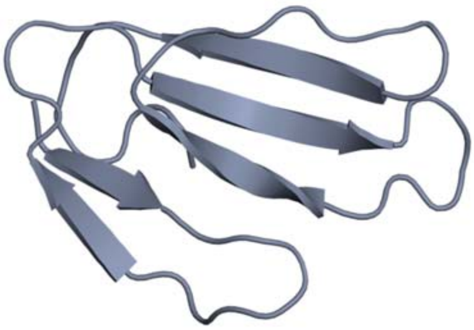

Fig. 1.1

a) Describe the molecular structure of fasciculin-2. [2]

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b) With reference to hemolytic toxins and neurotoxins, explain why snake venom, which has been heated to 100°C for several minutes, would likely lose its toxicity. [4]

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c) State how enzymes which hydrolyze phospholipids damage tissues. [1]

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d) Some antibodies bind to toxins and inactivate them. These antibodies are known as anti-toxins. The human immune response is far too slow to be effective in making anti-toxins against snake venom.

Injecting a very small, non-lethal quantity of venom into a horse produces anti-toxin. The horse produces anti-toxins that can be extracted from horse blood and used as an emergency treatment for those bitten by the same species of snake. Each time the horse is injected with venom, it is able to tolerate larger doses and the concentration of the specific anti-toxin in its blood is greater.

i) Explain why the human immune response is too slow to protect a person from a snake bite. [3]

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ii) Explain why a horse is injected more than once with a small amount of venom when it is being prepared for use as a source of anti-toxin. [2]

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iii) It was observed that upon injection with the toxin, different types of anti-toxins are produced. Each type of anti-toxin is different at the variable region, but they are equally effective against the toxin.

Suggest why different anti-toxins are produced. [2]

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iv) Explain why treatment with the horse anti-toxin will not produce long-term protection against snake bites. [3]

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e) In recent years, contortrostatin, a protein found in the venom of the southern copperhead snake, has been extensively demonstrated to hold great promises in cancer treatment. Contortrostatin binds to and disrupt the function of integrins, causing tissue damage. Integrins are transmembrane proteins that serve as bridges for cell-to-cell interactions, which is important in adhering endothelial cells of blood vessels to our body tissues.

Using the information above and your knowledge on the characteristics of cancer cells, suggest why contortrostatin can be used as a medicine in cancer treatment. [3]

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f) Mice and monkeys have been successfully immunised against several important infectious diseases using experimental DNA vaccines, in the form of plasmids. Plasmids are small circular DNA molecules.

During the 1990s, researchers found that mouse muscle and other mouse tissues were able to absorb plasmids which had been injected into the animals. Any genes that were part of this plasmid DNA were transcribed and translated. The resulting polypeptides were presented on the cell surface in complex with host receptor molecules, which allow the immune system to recognise the polypeptide as non-self. Proteins that are presented at the cell surface in this way stimulate the lymphocytes of the immune system very effectively.

This discovery allows plasmid DNA to be used as a vaccine, even though the DNA does not itself act as an antigen. Most vaccines contain proteins, or fragments of proteins, that are extracted from the surface of pathogens. It is a complex and costly procedure to purify these protein antigens.

Fig. 1.2 shows a simplified diagram of a DNA vaccine. This plasmid codes for two antigens, **A** and **B**.

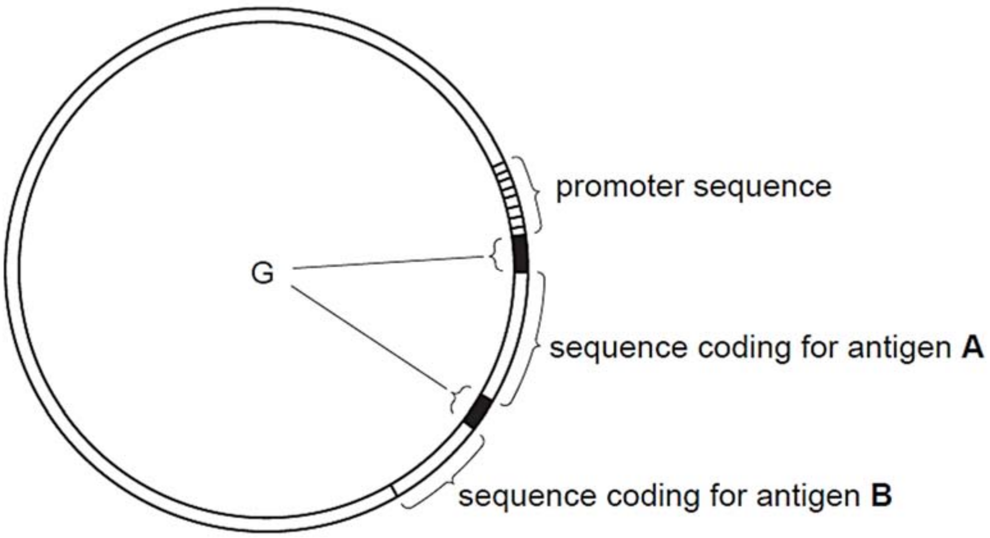

Fig. 1.2

i) Suggest why proteins presented at the cell surface of antigen-presenting cells are able to stimulate an immune response more effectively than proteins dissolved or suspended in the blood or tissue fluids. [1]

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ii) Sequences of nucleotides, labelled **G** on Fig. 1.2, code for groups of amino acids at the beginning of each polypeptide. These amino acid sequences direct the newly-synthesised polypeptides to the rough endoplasmic reticulum of the muscle cell.

Suggest how this makes the vaccine effective. [2]

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iii) Suggest why it may be advantageous to include nucleotide sequences coding for more than one antigen in a DNA vaccine. [1]

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[Total: 24]

QUESTION 2

In an attempt to control the spread of dengue, using genetic modification, a piece of DNA is inserted into the *Aedes aegypti* mosquito genome at the embryonic stage. This DNA contains a lethal gene (*tTAV* gene) which codes for a protein called tTAV. This protein acts as a molecular switch to shut down the expression of **all** other genes, leading to death of the insect. In order to shut down all other genes, the tTAV protein concentration in the cell must reach a high concentration. To achieve that, the tTAV proteins that are initially synthesized also function to increase the expression of its own gene, thereby producing even more tTAV proteins, demonstrating what is known as a positive feedback.

This tTAV protein, however, is inactivated by a compound called tetracycline, which is incorporated into the food that the developing larvae feed on. Hence, the genetically-modified (GM) larvae survive to adulthood, with much of the tetracycline still remaining in them. Male GM mosquitoes are then selected to breed with females to produce large number of offspring. The male GM offspring are selected and fed with tetracycline until they reach adulthood. They are then released into the wild to mate with wild-type females. Any offspring larvae produced will contain the *tTAV* gene, which is expressed to cause death of the larvae.

a) Explain how tTAV protein increases the expression of its own gene. [2]

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b) Suggest a reason why a high concentration of tTAV proteins would shut down the expression of **all** other genes. [2]

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c) Suggest **two** advantages of using GM mosquitoes over the use of pesticides in controlling the spread of dengue. [2]

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Male GM mosquitoes have been used in open field trials in countries such as Cayman Islands. The town where the *Aedes aegypti* mosquitoes predominate was divided into three areas, as shown in Fig. 2.1.

Area **A** – the treatment site where male GM mosquitoes are released
Area **B** – buffer zone
Area **C** – the non-treated control site

The mosquito populations in area **A** and area **C** were measured using an ovitrap – a device that is attractive as an egg-laying site for female mosquitoes.

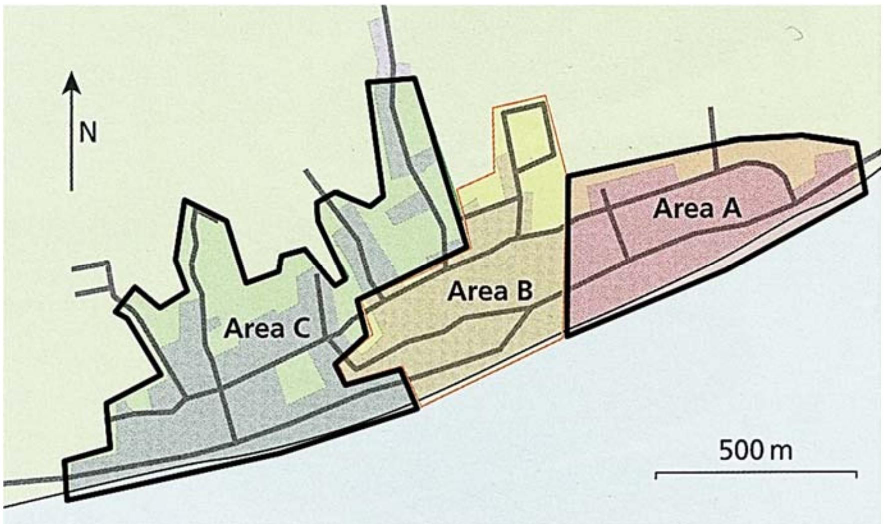

Fig. 2.1

d) State why the release of male GM mosquitoes in area **A** will not increase the risk of transmission of dengue. [1]

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e) Suggest the purpose of area **B**. [1]

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The number of ovitraps that contain eggs were recorded every week for 6 months. Fig. 2.2 shows the ovitrap index, which is calculated based on the percentage of ovitraps containing eggs.

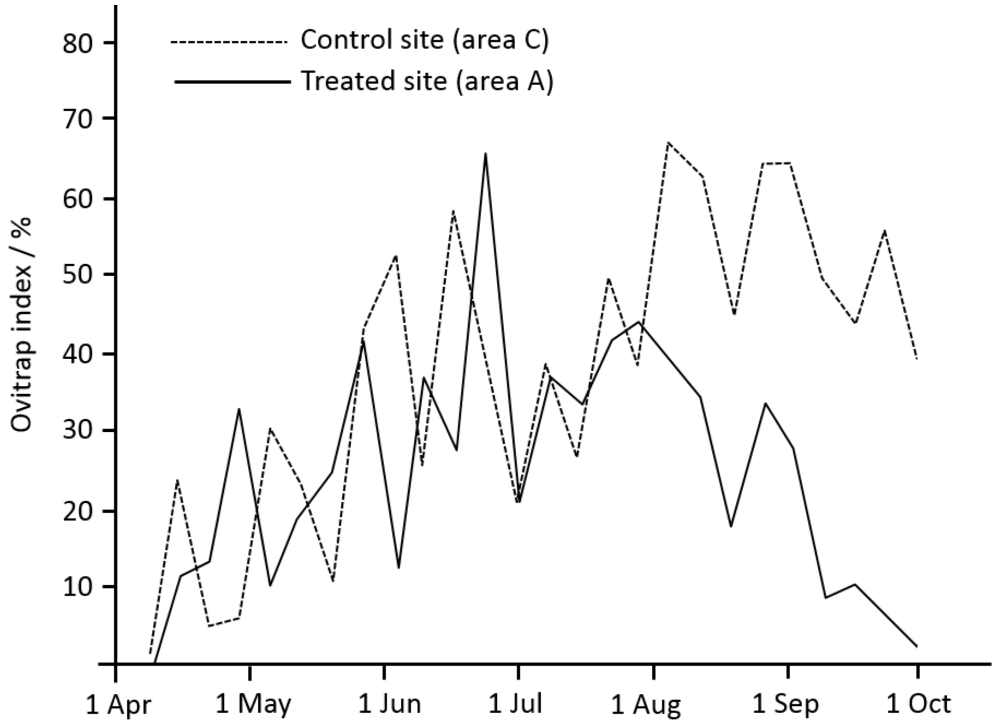

Fig. 2.2

f) Comment on the trend observed for both the treated and control site. [4]

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g) Global warming is the unusually rapid increase in Earth's average surface temperature over the past century primarily due to the greenhouse gases released as a result of anthropogenic activities. The global average surface temperature rose by 0.6 to 0.9°C between 1906 and 2005, and the rate of temperature increase has nearly doubled in the last 50 years. Temperatures are certain to go up further.

i) Explain how global warming has encouraged the spread of dengue. [3]

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ii) Apart from the spread of mosquito-borne diseases, global warming is already putting pressure on ecosystems (the plants and animals that co-exist in a particular climate zone), both on land and in the ocean. Warmer temperatures have already shifted the growing season in many parts of the globe. Spring is arriving earlier in both hemispheres, causing the growing season in parts of the Northern Hemisphere becoming two weeks longer in the second half of the 20th century.

This change in the growing season also affects the broader ecosystem. Migrating animals have to start seeking food sources earlier. Furthermore, the shift in seasons may already be causing the life cycles of pollinators, like bees, to be out of sync with flowering plants and trees. This mismatch can limit the ability of both pollinators and plants to survive and reproduce, which would reduce food availability throughout the food chain.

Describe how global warming has impacted **other** biotic factors. [3]

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[Total: 18]

QUESTION 3
Ribulose bisphosphate carboxylase-oxygenase, better known as rubisco, is a massive protein made up of 16 subunits, and is an important enzyme that all life forms depend on. It has a low affinity for carbon dioxide and fixes only 3–10 molecules of carbon dioxide per second, compared to other enzymes which convert hundreds to millions of substrates per second. The consequence is that photosynthetic cells synthesize a large amount of rubisco. About half of all proteins in green leaves consist of rubisco, making this enzyme the world's most abundant protein.

In addition to carbon dioxide, the same active site of rubisco that binds carbon dioxide also binds oxygen, hence this enzyme has 'oxygenase' in its name. The oxygenase activity of rubisco combines oxygen to RuBP, which is split into 3-phosphoglycerate and a two-carbon compound called 2-phosphoglycolate, as shown in Fig. 3.1.

2-phosphoglycolate is converted into 3-phosphoglycerate in a series of reactions, but this pathway consumes oxygen and releases carbon dioxide, hence the name *photorespiration* is given to this pathway. The rate of photorespiration is usually about one-third that of the Calvin cycle, but this rate is predicted to increase with global warming, reducing plant productivity.

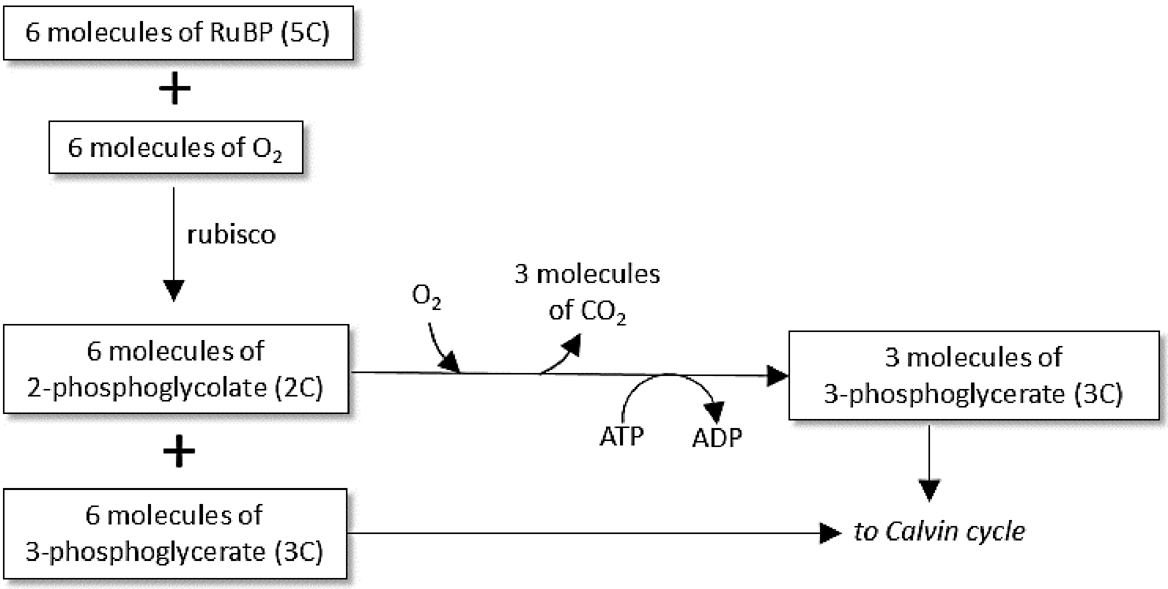

Fig. 3.1

a) Explain why all life forms are dependent on rubisco. [3]

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b) Explain why photorespiration will reduce the rate of the Calvin cycle. [2]

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c) Rubisco has been described as a classic example of '*unintelligent evolutionary design'*.

Explain why. [3]

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[Total: 8]

Section B

Answer **ONE** question in this section.
Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 4

a) DNA molecules replicate with a high degree of accuracy, yet not always perfectly.

Describe how this occurs and discuss why the survival of a species depends on DNA molecules being stable, yet not *absolutely* stable. [13]

b) Discuss the view that all life forms depends on phosphate. [12]

[Total: 25]

QUESTION 5

a) It is observed that cancer cells share similar characteristics as stem cells, yet there are characteristics which distinguish them.

Describe the above observations and explain the molecular basis of cancer. [13]

b) Discuss the importance of hydrogen bonding in ensuring the continuity of life. [12]

[Total: 25]

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☺ **End of Paper 3** ☺